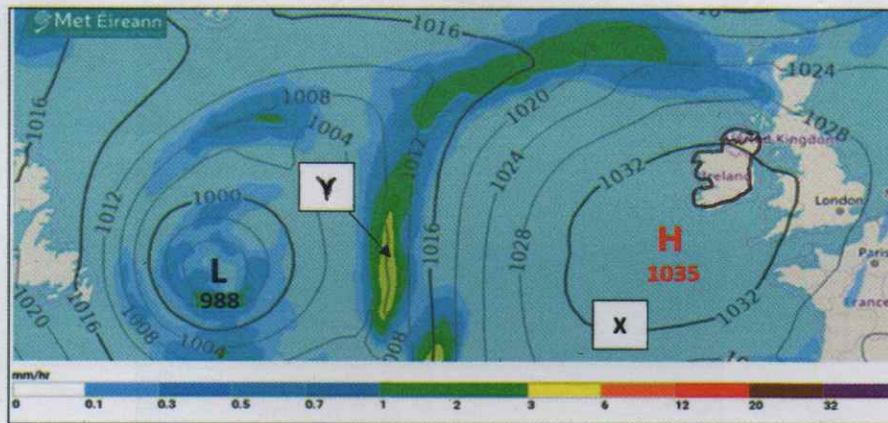


Name: Oisín Argand

Question 1

- (b) Examine the Met Éireann weather chart below and tick (✓) the correct box for each of the following questions.



- (i) Which of the following is Ireland under the influence of?
- High Pressure ☒ 3
- Low Pressure ☐
- (ii) What is the atmospheric pressure over the area marked X?
- 1028 mb ☐
- 1024 mb ☐ 3
- 1032 mb ☒
- (iii) On the chart above, in which area of pressure will there be higher precipitation?
- High Pressure ☐ 3
- Low Pressure ☒
- (iv) Which of the following levels of precipitation is forecast to fall in the area marked Y?
- 0.3 mm – 0.5 mm ☐
- 3 mm – 6 mm ☒ 3
- 12 mm – 20 mm ☐
- (v) Which of the following is the unit of measurement for precipitation?
- Millimetres (mm) ☒ 3
- Millibars (mb) ☐
- (vi) Which of the following is the instrument used to measure precipitation?
- Barometer ☐
- Rain gauge ☒ 3
- Anemometer ☐

18

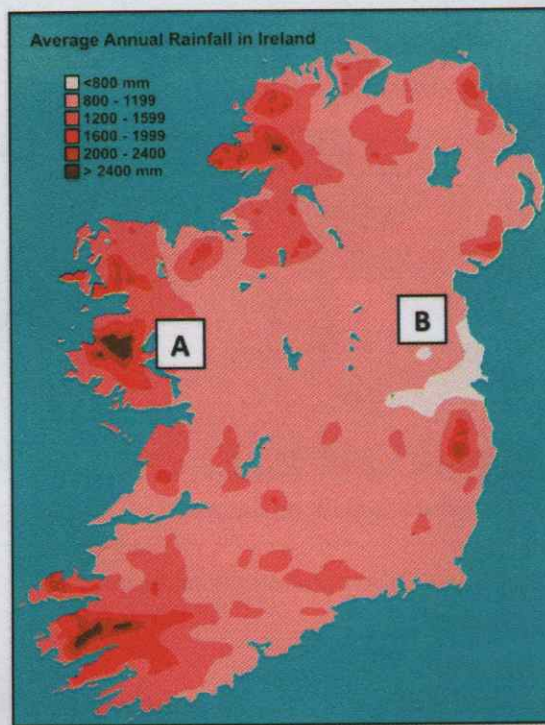
144 / 170 = 85%

Well done
Oisín

Question 2

Question 3

- (a) Examine the rainfall chart below and answer each of the following questions.



- (i) Which of the following counties receives the highest amount of rainfall?
Tick (✓) the correct box.

Dublin ☐

Clare ☐

Kerry ☒

- (ii) Lines on a map showing areas of equal rainfall are called Isohels. Is this statement true or false?
Tick (✓) the correct box.

True ☒

False ☐

- (iii) At which location on the rainfall chart is relief rainfall most likely to occur?
Tick (✓) the correct box.

A ☒

B ☐

3

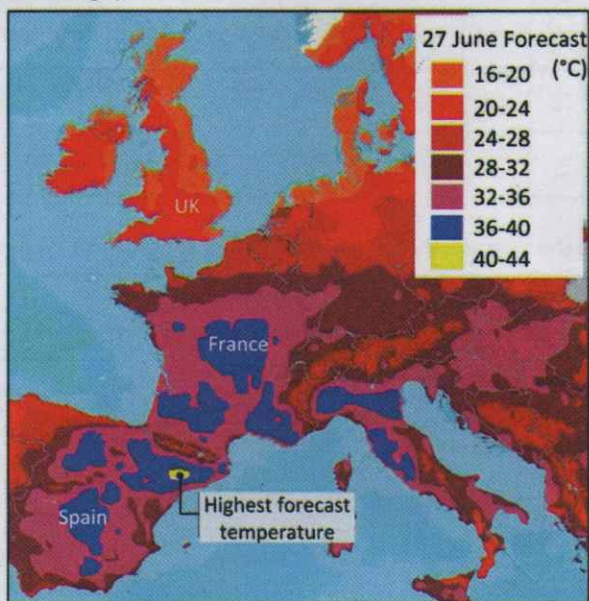
3

6

Question 3

Question 5

- (a) Examine the map below showing temperature ranges forecast in Europe for 27 June 2023 and answer the following questions.



- (i) What was the lowest forecast temperature in Ireland?
Tick (✓) the correct box.
- 24-28°C ☐ 28-32°C ☐ 16-20°C ☒
- (ii) Which of the following countries has the highest forecast temperature?
Tick (✓) the correct box.
- Spain ☒ France ☐ UK ☐
- (iii) Name Ireland's climate.

Moderate oceanic climate *Cool temperate oceanic*

- (iv) Explain **one** reason why areas in northern Europe are more likely to have lower temperatures than areas in southern Europe.

Southern Europe is closer to the equator. This means it receives more direct sunlight than Northern Europe, which receives indirect sunlight at an angle.

17

Question 4

- (b) The table below shows the climate data for Santorini, a popular tourist destination in Greece that has many beaches. Study the table and answer each of the following questions.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Temperature (°C)	12	13	14	16	20	24	26	26	24	21	17	13
Rainfall (mm)	71	43	40	16	11	0	7	0	11	38	60	73
Average Daily Sunshine (hours)	7	7	9	11	12	13	14	13	11	9	8	6

- (i) What was the average daily sunshine (hours) in Santorini for April?

11 hours

- (ii) What is the annual temperature range in Santorini?

Tick (✓) the correct box.

10°C ☐

12°C ☐

14°C ☒

- (iii) Calculate the mean (average) rainfall for the three months of October, November and December. Use the correct unit of measurement in your answer.

Optional rough work

60 + 73
133
+ 38
171

171 mm — 57 mm

- (iv) Most tourists visit Santorini in the months of June, July and August. Explain one reason why these months are attractive for tourists to visit this area. Use evidence from the table to support your answer.

The weather is at its best within these months. There is almost no rainfall (0mm in June and August, 7mm in July). The weather is the warmest (gets 24°C in June and 26°C in July and August) and sunshine hours are at their longest (13 hours in June and August, 14 hours in July).

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Question 5

11. METEROLOGICAL INSTRUMENTS

Circle the correct answer in each of the following statements.

- (i) An instrument used to measure levels of rainfall is known as a gauge / barometer.
- (ii) A Campbell Stokes recorder is used to measure levels of temperature / sunshine.
- (iii) Humidity is recorded by using a hygrometer / thermometer.



Question 6

Question 6

- (a) (i) Use the weather instruments in the box below to fill in the table provided.

Barometer	Anemometer	Hygrometer	Rain Gauge
Weather Instrument		Unit of Measurement	
Anemometer		Km/per hour	
Rain gauge		Millimetres (mm)	
Barometer		Hectopascals (hpa)	
Hygrometer		Percentage	

(ii)



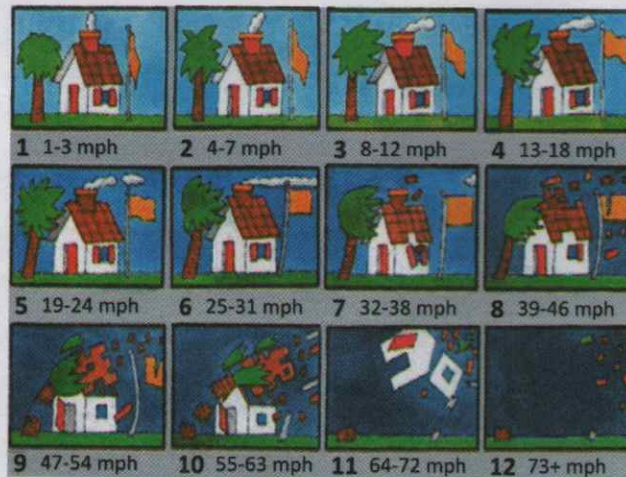
On a weather chart, the symbol above indicates which of the following?
Tick (✓) the correct box.

Warm front ☐

Occluded front ☐

Cold front ☒

(iii)



Which of the following terms is used to describe a scale that measures the strength of wind?

Tick (✓) the correct box.

Beaufort scale

☒

Simpson-Saffir scale

☐

(iv)



The weather conditions in the photograph above are associated with which of the following forms of atmospheric pressure?

Tick (✓) the correct box.

High pressure ☒

Low pressure ☐

3

(b) (i) Name **one** example of a significant weather event you have studied.

Hurricane Ian

2

(ii) When did the weather event happen?

Late September
2022

~~Western Cuba and Southern USA~~

2

(iii) List **two** effects of the significant weather event that you have studied.

1. Loss of life, the death toll was 148 people
2. Destruction of property, it was the most destructive hurricane to make landfall in the US since the 1930s

4

25

- (iv) With the help of a labelled diagram, explain how the weather event you named in part (i) was formed.

Water was heated by the warm sun near the equator. It rose into the air as steam, where low pressure condensed it into clouds. The low pressure made it funnel upwards into a tower with wind speeds up to 119 km/h. As it moved over water, it got larger. It only shrank over land.

Space for diagram(s)

- (v) Name and explain any two ways the destruction caused by the weather event you named in part (i) could be reduced.

1. Better, more resilient infrastructure could reduce the damage to property. For instance, putting steel beams in concrete used in buildings.
2. An early warning system created by meteorologists could reduce loss of life by telling people to evacuate ahead of time.

Question 7

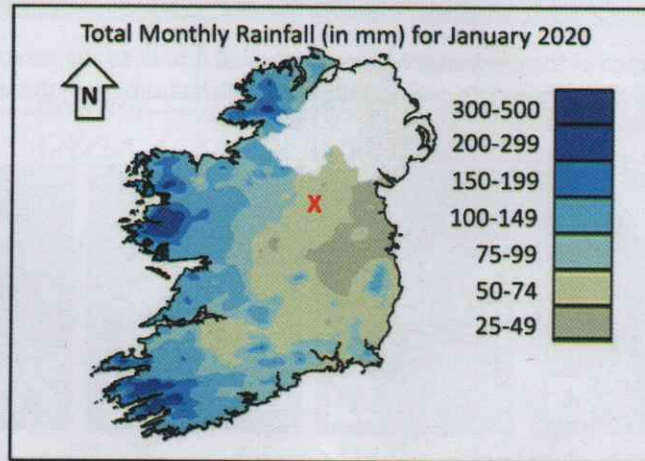
- (a) (i) Match each of the photographs below, labelled A to D, to the weather instrument that matches it by writing the correct letter for each instrument in the table below.



Weather Instrument	Letter
Wind/weather vane	C
Campbell-Stokes recorder	B
Rain gauge	D
Anemometer	A

17

Study the map of Ireland below showing the total monthly rainfall in Ireland for January 2020 and answer each of the following questions.



- (ii) The area marked X on the map experienced 200-299 mm of rainfall in January 2020. Tick (✓) the correct box.

True ☐

False ☒

- (iii) The weather map above shows that the highest levels of rainfall were along the west coast of Ireland.

Explain **one** reason why areas along the west coast of Ireland have more rainfall.

In warm areas near the equator, water is heated and forms clouds. The Mexican Gulf Stream pushes those clouds over the Atlantic, with many of them ending up in Ireland. As clouds move over land, they shrink. So rain clouds that arrive in the west of Ireland, often are much less potent by the time they reach the east, making the west experience more rainfall. Relief rain

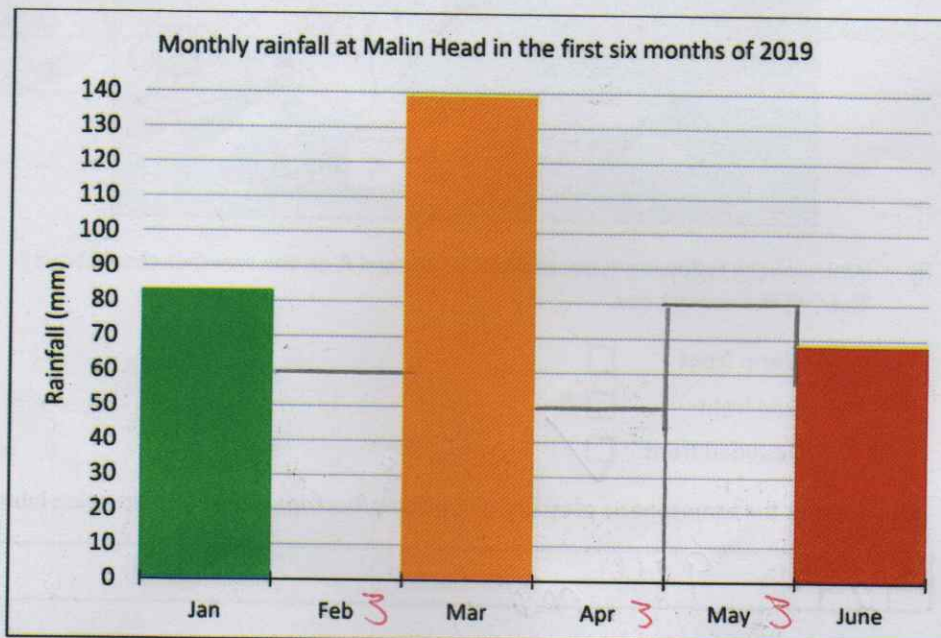
Question 8

- (b) Examine the table below showing monthly rainfall at Malin Head in the first six months of 2019 and answer each of the following questions.

Monthly rainfall at Malin Head in the first six months of 2019

	Jan	Feb	Mar	Apr	May	June
Rainfall (mm)	83	60	139	50	80	68

- (i) Complete the bar chart below using the information from the table for February, April and May.



- (ii) How much rain fell in March 2019 at Malin Head? Use the correct unit in your answer.

139 mm

- (iii) In which month did the lowest amount of rain fall at Malin Head?

April

- (iv) Calculate the mean monthly rainfall at Malin Head for the first six months of 2019. Use the correct unit in your answer.

Optional rough work:

Answer: 80 mm

$$\begin{array}{r}
 10 \overline{) 139} \\
 \underline{10} \\
 39 \\
 \underline{30} \\
 90 \\
 \underline{90} \\
 0
 \end{array}$$

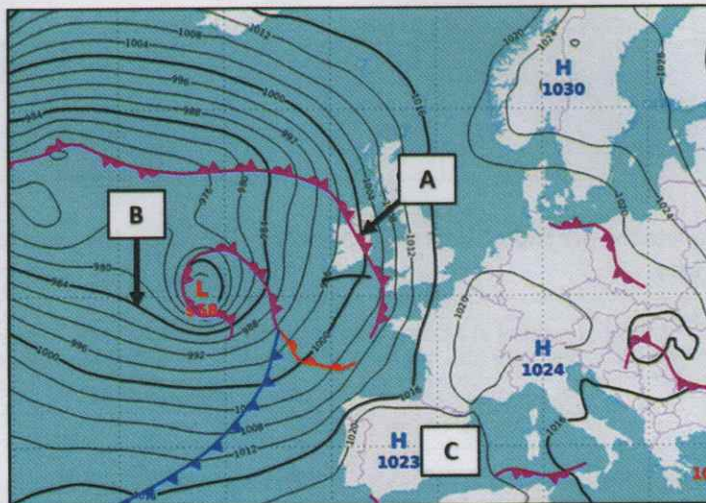
$$\begin{array}{r}
 19 \\
 \times 12 \\
 \hline
 38
 \end{array}$$

$$\begin{array}{r}
 6 \overline{) 480} \\
 \underline{36} \\
 120 \\
 \underline{120} \\
 0
 \end{array}$$

26

Question 9

- (c) Examine the weather chart below and answer each of the following questions.



- (i) Which of the following types of front is labelled A on the weather chart above?
Tick (✓) the correct box.

Warm front ☐
Cold front ☐
Occluded front ☒

- (ii) What is the atmospheric pressure (in millibars/hectopascals) at the location labelled B?

964 984 mb

- (iii) Atmospheric pressure is measured using an instrument called a barometer.

Is this statement true or false?

Tick (✓) the correct box.

True ☒
False ☐

- (iv) At which location shown on the weather chart can an anticyclone be seen?

Tick (✓) the correct box.

B ☐
C ☒

- (v) Using the weather chart on the previous page, contrast the weather conditions you would expect to find in the areas labelled A and C. Refer to both wind and precipitation in your answer.

In C, winds would be warm.
In A, winds would be mixed, but colder than C. In C, there would be a chance of precipitation, but more likely as rain or mist. In A, precipitation is more likely to be rain, but possibly also hail or snow.

8

8

17